Gerber Centrifuge



Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Gerber Centrifuge

What it is used for

To use a Gerber centrifuge to separate the fat layer from prepared milk samples during dairy fat testing. To obtain consistent separation by balancing tubes and following the instrument's operating instructions.

Main parts

- Centrifuge body and lid
- Rotor or sample tube holders
- Gerber centrifuge tubes
- Lid lock and safety interlock
- Speed and timer controls
- Brake and display, if fitted

Before you start

1. Place the centrifuge on a stable, level surface with adequate ventilation.
2. Inspect the rotor, tube holders, lid, and tubes for cracks, corrosion, or damage.
3. Confirm that the rotor and tubes are approved for the instrument and test method.
4. Prepare milk samples and Gerber tubes according to the laboratory procedure.
5. Balance opposing tubes by equal mass and use compatible tubes in matching positions.
6. Check that the lid lock and safety interlock function correctly.

How to use

1. Switch off the centrifuge before loading samples.
2. Place each tube in its holder and arrange balanced loads opposite one another.
3. Ensure all tubes are seated correctly and their closures are secure as required by the method.
4. Close the lid fully and confirm that the lock engages.
5. Set the speed, temperature if available, and running time specified by the test method and instrument manual.
6. Start the centrifuge and observe it from a safe distance for unusual vibration or noise.
7. Stop the run immediately if excessive vibration, unusual noise, or leakage occurs and wait for the rotor to stop completely.
8. When the cycle ends, wait until the rotor has stopped and the lid unlocks.
9. Open the lid carefully and remove tubes without disturbing the separated layers.
10. Read the fat column using the approved Gerber method and record the result.
11. Switch off and clean any contamination according to laboratory procedures.

⚠ Safety

- Use only tubes, stoppers, and rotors rated for the selected speed and temperature.
- Never open the lid or attempt to slow the rotor by hand while it is moving.
- Do not operate with cracked tubes, damaged holders, or an unbalanced load.
- Wear a lab coat, safety glasses, and suitable gloves when handling chemicals and samples.
- Follow the approved method for handling sulfuric acid, amyl alcohol, and hot tubes where applicable.
- Disconnect power before maintenance and allow hot components to cool.

Care & storage

- Clean the chamber and holders after use using a method approved by the manufacturer.
- Dry components thoroughly and check for corrosion or residue.
- Inspect the rotor, lid seal, and locking mechanism regularly.
- Keep ventilation openings clear and arrange periodic servicing.
- Record faults and remove damaged components from service.

Common mistakes to avoid

- Loading tubes without balancing their mass.
- Using tubes or a rotor that are not rated for the operating conditions.
- Opening the lid before the rotor stops completely.
- Using incorrect test settings or an unapproved sample preparation method.
- Leaving corrosive residue or sample spills inside the chamber.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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